

2.8 Proofs – Part 2

Lesson Introduction

 Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	 Learning Targets	- I can reflect on the mathematical thinking found in a proof. - I can reason about proofs using deductive reasoning, definitions, and postulates.	
 Benchmark Coherence	Building On MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.	Working Toward N/A		
 Essential Questions	- What are the benefits of reflecting on the mathematical thinking found in a proof? - How can I use deductive reasoning, definitions, and postulates to reason about proofs?			
 Lesson Narrative	In this lesson, students will be moving into the second lesson of a two-lesson series on proofs. Students will be completing proofs from start to finish. They will observe and then participate in the process of brainstorming before writing a proof. Students will complete a flowchart proof and then translate that proof into a two-column proof format. This lesson focuses on the use of MA.K12.MTR.4.1 and is focused more on the learning targets than the actual content of the benchmark. It is very important for students to understand these ideas to master the content in geometry.			
 Component Summary	2.8.1 Warm-Up (~10 min.)	Bell Work writing a paragraph proof (<i>SE p. 117</i>)	2.8.3 Guided Instruction (10 min.)	Brainstorming and writing a flowchart proof (<i>SE p. 119</i>)
	2.8.2 Collaboration (15 min.)	Examining brainstorming before writing a proof (<i>SE p. 117</i>)	2.8.4 Your Turn! (10 min.)	Translating a flowchart proof to a two-column proof (<i>SE p. 120</i>)
	2.8.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 121</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Paragraph Proof - Flowchart Proof	N/A	N/A	

 Teacher Tips	Common Misconceptions • Students may forget to use a given statement as part of their original proof. • Students may skip a step in the proof process. • Students may get lost in the flow of a flowchart proof. • Students may make assumptions instead of using appropriate reasons.	Things to Consider • Consider helping students understand the importance of brain storming before starting a proof. Establish a culture within your class where students can freely share their ideas and to discuss their thinking with their peers and with you. • Consider how allowing students to struggle and problem solve helps them deepen their thinking. Refrain from “rescuing” students too early or rushing students through their productive struggle. • Consider using information from the previous lessons to form a small group of students who are struggling working with proofs. Facilitate these students as they work through 2.8.4 Your Turn! • If time is an issue, then consider having students finish the proof in 2.8.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Correct 2.8.2 Collaboration has students work together to examine the process of brainstorming before writing a proof. After reviewing two examples of brainstorming notes, students can engage in this protocol to clarify, critique, and correct the notes as part of the activity.	MA.K12.MTR.4.1: Discussion and Reflection 2.8.2 Collaboration is an opportunity for students to continue exploring proofs. Students will engage in collaborative discussions about proofs and brainstorming. They will be expected to analyze the mathematical thinking of others and communicate mathematical ideas, vocabulary, and methods effectively.
 Differentiate Instruction	This lesson will be the first experience students have with writing proofs from scratch. Consider providing students with a word bank to choose from that they could match to the corresponding sections of the proofs.	
Lesson Notes:		

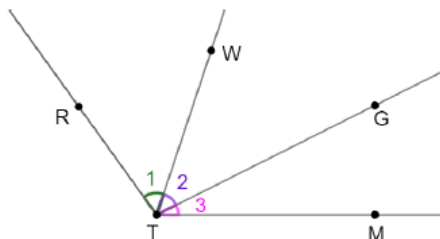
2.8.1 Warm-Up: Bell Work

Component Overview: Students will write a paragraph proof from the flowchart proof and two-column proof from the previous lesson. Students should complete this activity independently



2.8.1 Warm-Up: Bell Work

Given: $m\angle 1 = m\angle 3$
 Prove: $\angle RTG \cong \angle WTM$



1. Use the flowchart proof and the two-column proof from the previous lesson to write a paragraph proof.

It is given that $m\angle 1 = m\angle 3$. Using the addition property of equality, we know that $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$.

Using the angle addition postulate we know that $m\angle 1 + m\angle 2 = m\angle RTG$ and that $m\angle 2 + m\angle 3 = m\angle WTM$.

Next we apply the substitution property of equality to determine that $m\angle RTG = m\angle WTM$. By the definition of congruence, we reach our conclusion that $\angle RTG \cong \angle WTM$.



For this activity, consider reviewing the instructions with the students prior to having them work on this question. For timing purposes, it is important that students know that this is the same question that they wrote the flowchart and two-column proof for in the prior lesson. If students begin trying to write the proof from scratch, it may take more time than desired. Allow students to look back to proofs from Lesson 7.

2.8.2 Collaboration: Brainstorming Before Writing a Proof

Component Overview: Students will work collaboratively to examine two examples of brainstorming before writing a proof. For this activity, students should be allowed to work with a partner or small group.

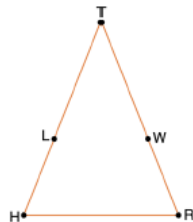


2.8.2 Collaboration: Brainstorming Before Writing A Proof

Work with your partner to complete the following.

Gabrielle and Samuel were given the following information.

$\triangleTHR$ is shown where $\overline{HL} \cong \overline{WR}$,
 $\overline{LT} \cong \overline{WT}$.
 Prove that $\overline{HT} \cong \overline{TR}$.



Before Gabrielle and Samuel wrote their paragraph proof, they brainstormed. Their notes are shown below.

Gabrielle



Given: $\overline{HL} \cong \overline{WR}$ Prove: $\overline{HT} \cong \overline{TR}$
 $\overline{LT} \cong \overline{WT}$

$HT = HL + LT$ adding two segments to get the total
 What is that called?

If using = then it has to be changed to congruence.

Samuel



Given: $\overline{HL} \cong \overline{WR}$ $\overline{LT} \cong \overline{WT}$
 Prove: $\overline{HT} \cong \overline{TR}$ don't worry, it's the same as $\overline{RT}$

Basically if you add the two segments of each side you get the entire side.

- Why did Samuel write "Don't worry it is the same thing as $\overline{RT}$?"
 Samuel wrote "Don't worry it is the same thing as $\overline{RT}$ " because $\overline{TR}$ and $\overline{RT}$ represent the same segment between the same two points. When naming a segment, it does not matter which endpoint you name first.



This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Teachers can summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 2.8.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.



For this activity, consider pairing or grouping the students based on their performance in the prior lesson. This activity is directly connected to the concept of writing a proof, which was introduced in the prior lesson.

Consider using high/low combinations based on performance to allow students to work together through the concept of brainstorming before writing a proof while reinforcing the concepts of proofs that were introduced in the prior lesson.



If students have been making a proof resource, encourage students to look through and use their resource to find the names of any definitions, postulates, and theorems.

2.8.2 Collaboration: Brainstorming Before Writing a Proof (continued)

2. Gabrielle made some markings on the triangle she drew. What are her markings meant to show?

The two segments with single dashes are congruent ($\overline{HL} \cong \overline{WR}$), and the two segments with double dashes are congruent ($\overline{LT} \cong \overline{WT}$).

3. Gabrielle wrote " $HT = HL + LT$," but she could not remember what that was called. Samuel wrote "Basically if you add the two segments of each side you get the entire side." What postulate justifies Gabrielle's and Samuel's statements?

If you and your partner cannot remember the name of the postulate Gabrielle and Samuel are referring to, then look through your workbook to find it.

The postulate that Gabrielle and Samuel are referring to is the segment addition postulate.

4. Gabrielle only wrote " $HT = HL + LT$," but Samuel mentioned "each side." Write the equation for the other side of the triangle. $RT = RW + WT$

5. Gabrielle wrote "If using $=$ then it has to be changed to congruence." What definition justifies changing from equality to congruence?

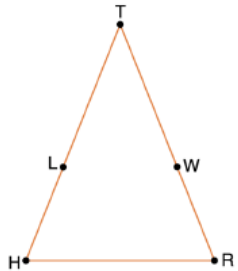
The definition of congruence.



While circulating pay particular attention to notations when showing examples of the proof for question 6. Students need to understand the importance of proper notation. For example, measures are added, not the segments themselves, so the segment notations are not used for the equations but are used when stating congruence.

2.8.2 Collaboration: Brainstorming Before Writing a Proof (continued)

6. Apply the brainstorming of Gabrielle and Samuel to finish the paragraph proof. The first sentence has been written for you.



Given: $\overline{HL} \cong \overline{WR}$, $\overline{LT} \cong \overline{WT}$

Prove: $\overline{HT} \cong \overline{RT}$

It is given that $\overline{HL} \cong \overline{WR}$ and $\overline{LT} \cong \overline{WT}$. By definition of congruence, $HL=WR$ and $LT=WT$. $HL+LT = HT$ and $WR+WT = RT$ by segment addition postulate. Since $HL=WR$ and $LT=WT$, then by substitution property of equality $WR+WT = HT$. Therefore, by the transitive property of equality $HT=RT$. In conclusion $\overline{HT} \cong \overline{RT}$ by the definition of congruence.

7. Describe the most difficult part of writing the paragraph proof.
Answers will vary.
8. Describe how Gabrielle and Samuel's brainstorming was helpful.
Answers will vary.
9. Check your proof with another group and if necessary, resolve any differences.



For question 8, consider having student pairs or groups check with at least two other groups and have each group review and initial that they agree with the group findings. This will allow for multiple perspectives and physical movement in this activity.

2.8.3 Guided Instruction: Flowchart Proof

Component Overview: Students will first brainstorm and then write a flowchart proof. For this activity, after allowing students to try question 1, have groups share their brainstorming prior to writing the flowchart proof to check for understanding. Then guide students through question 2.



2.8.3 Guided Instruction: Flowchart Proof

$\overline{AX}$ and $\overline{DY}$ are shown.



Given: K is the midpoint of $\overline{AX}$, L is the midpoint of $\overline{DY}$, and $\overline{AX} \cong \overline{DY}$

Prove: $\overline{AK} \cong \overline{DL}$

1. Brainstorm with your partner about how to write the proof. Then, write your brainstormed ideas in the space provided. Include any possible definitions or postulates that may need to be used to justify the steps.

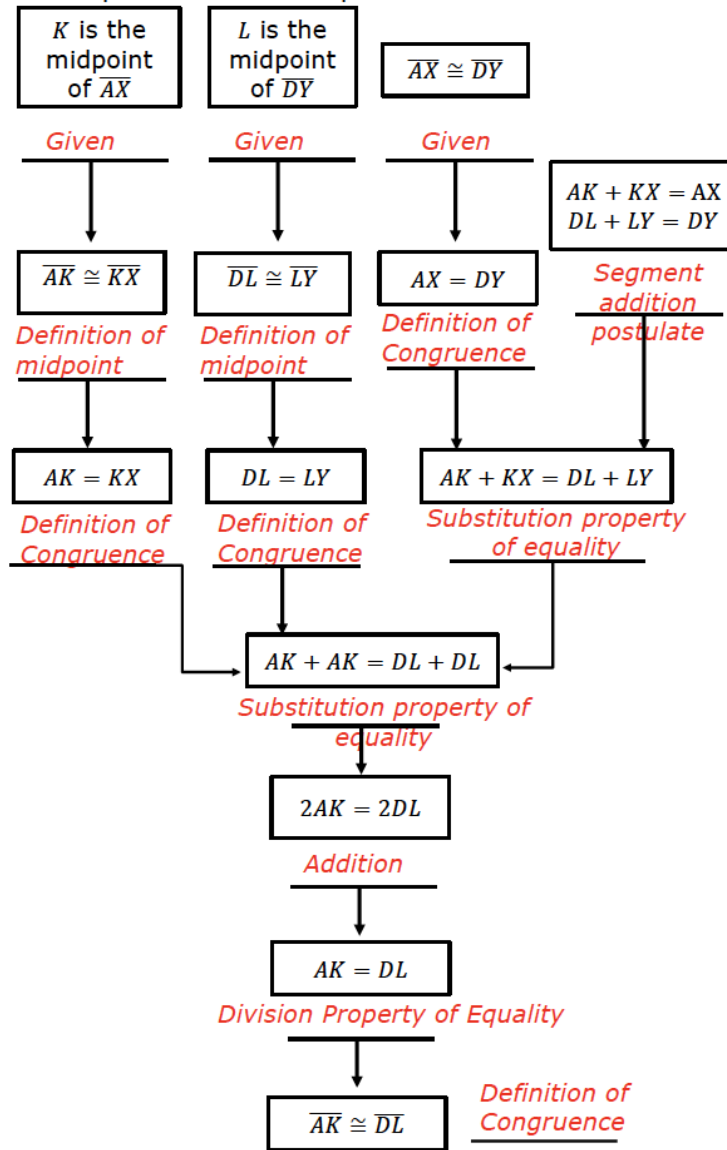
Answers will vary. We discussed needing the definition of bisect. We might also need the segment addition postulate and the definition of congruence.



This proof has a higher cognitive demand. Consider monitoring the length of time that students struggle and help support them where needed. Students may have the misconception that all proofs are direct, which is not the case.

2.8.3 Guided Instruction: Flowchart Proof (continued)

2. Complete the flowchart proof.



Consider creating choice cards as part of this question that students can choose from to complete the boxes if that level of scaffolding is needed.



If students are ready for a challenge, have students close their books and attempt this flowchart proof without the use of a template to see how they approach the problem from start to finish.

2.8.4 Your Turn!

Component Overview: Students will use the flowchart from the prior activity to complete the two-column proof. For this activity, students should work independently to practice what they've been learning in this lesson.



2.8.4 Your Turn!

1. Use the previous flowchart to complete the two-column proof.

Statement	Reason
1. K is the midpoint of $\overline{AX}$ and L is the midpoint of $\overline{DY}$	1. Given
2. $\overline{AK} \cong \overline{KX}$ and $\overline{DL} \cong \overline{LY}$	2. Definition of midpoint
3. $AK = KX$ and $DL = LY$	3. Definition of congruence
4. $\overline{AX} \cong \overline{DY}$	4. Given
5. $AX = DY$	5. Definition of congruence
6. $AK + KX = AX$ and $DL + LY = DY$	6. Segment addition postulate
7. $AK + KX = DL + LY$	7. Substitution property of equality
8. $AK + AK = DL + DL$	8. Substitution property of equality
9. $2AK = 2DL$	9. Addition
10. $AK = DL$	10. Division property of equality
11. $\overline{AK} \cong \overline{DL}$	11. Definition of congruence



This activity can be modified to be scaffolded based on student needs. For example:

- If students are understanding the concept very well, consider having them complete the proof without the table in the book from start to finish
- If students are struggling with this concept, consider the use of choice cards to put together pieces of the puzzle in this process
- If students are doing fairly well, leave the activity unmodified and do it as is.

2.8.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



2.8.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
Answers will vary.



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.